

Supporting Information for

“Separating the known from the unknown at Marfa, Texas: line of sight, refraction, and photometry of the ordinary lights seen from the Marfa lights viewing area”

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This document gives the full methods behind the main paper, additional figures and tables, and a guide to the companion website. Equation, figure, table, and section numbers without an “S” refer to the main paper. All numbers are produced by the code in the project repository (<https://github.com/zacharyslate/marfa-lights-investigation>); the file that holds each result is named in the text.

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S1 Data and provenance

The terrain and geoid files are pinned in `data/dem/MANIFEST.json` with their source URLs, sizes, SHA-256 checksums, and the servers’ last-modified dates; the point-cloud tiles used are listed in `data/dem/ept_nodes.json`, and the other inputs are stored in `data/inputs/`. Table S1 lists them.

Table S1: Input data.

Layer	Product	Use
Terrain, primary	USGS 3D Elevation Program 1-m DEM, lidar project TX_WestTexas.2018.D19; 27 tiles of 10×10 km; quality level 2; Optech Galaxy sensor; collected February–May 2019	every terrain sample inside the tiles
Terrain, fallback	USGS 1/3 and 1 arc-second DEMs (editions of 30 August 2022); 4 and 14 tiles	outside the lidar; skyline to 120 km
Lidar accuracy	project check-point test: non-vegetated RMSE_z 5.6 cm (565 points), vegetated 95th percentile 21.6 cm (461 points)	terrain error model
Point cloud	the same lidar as Entwine Point Tiles (work units B3 and B7); 894 octree nodes, 29.4 million points around the grazing corridors	obstructions
Geoid	NOAA GEOID12B (the model used to produce the lidar heights); GEOID18 for sensitivity	NAVD88 $\rightarrow$ ellipsoidal heights
Roads	TxDOT Roadways, extract of 1 September 2026; 821 public-road features	lamp positions
Traffic	TxDOT annual average daily traffic (AADT), 2025 and two previous years	occupancy and rates
Railroads	USDOT/BTS North American Rail Network lines; FRA highway–rail crossing inventory (night-time through trains)	locomotive positions and rates
Towers	FCC Antenna Structure Registration, with registered lighting	fixed lights
Headlamps	UMTRI market-weighted beam patterns: low beam UMTRI-2004-23 Table 4; high beam UMTRI-2001-19 Table 6	intensity
Signal lamps	FMVSS No. 108 photometric requirements as tabulated in NHTSA TP-108-13	intensity brackets
Air clarity	National Park Service, Big Bend standard visual range	extinction

Observer. The eye is 1.6 m above the viewing platform at 30.2751108°N , $103.8827973^\circ\text{W}$. The lidar resolves the platform as a pad 1.3–1.5 m above the surrounding ground, with its surface at 1494.80 m NAVD88; the 1 arc-second DEM, which does not resolve it, gives 1493.57 m. The geoid undulation there is -21.19 m.

Road centerline. A hand-traced US-67 line used in an earlier version differs from the TxDOT centerline by 2.0 m (median), 4.9 m (90th percentile), and 9.3 m (maximum).

S2 Line-of-sight geometry

S2.1 Coordinates

Each point is converted from geodetic longitude, latitude, and ellipsoidal height $h = H + N$ (orthometric height plus geoid undulation) to Earth-centered, Earth-fixed Cartesian coordinates on the WGS84 ellipsoid. Horizontal paths are geodesics, computed with Karney’s algorithm[1] (the `pyproj` implementation). The terrain between the eye O and the lamp T is sampled along the geodesic.

S2.2 Chord frame and the critical coefficient

Let $\hat{\mathbf{u}}$ be the unit vector from the eye O to the lamp Q and $\hat{\mathbf{w}}$ the unit vector perpendicular to it in the plane spanned by $\hat{\mathbf{u}}$ and the mean ellipsoid normal. For each terrain sample P_i ,

$$s_i = (\mathbf{P}_i - \mathbf{O}) \cdot \hat{\mathbf{u}}, \quad y_i = (\mathbf{P}_i - \mathbf{O}) \cdot \hat{\mathbf{w}}. \quad (\text{S1})$$

With a constant refraction coefficient the ray is an arc lying above the chord by $\sigma_i(k) = ks_i(D - s_i)/2R$ (Eq. 2 and Appendix A), where R is the radius of curvature of the normal section of the ellipsoid in the azimuth α of the sight line, from Euler’s formula,

$$\frac{1}{R} = \frac{\cos^2 \alpha}{R_M} + \frac{\sin^2 \alpha}{R_N}, \quad (\text{S2})$$

with R_M and R_N the meridional and prime-vertical radii of curvature at the mean latitude. The lamp is visible if and only if $k \geq k_{\text{crit}} = \max_i 2Ry_i/[s_i(D - s_i)]$ (Eq. 3). The clearance at k is $c(k) = \min_i [\sigma_i(k) - y_i]$, and the apparent elevation of the lamp at the eye is the chord elevation plus $kD/2R$.

S2.3 Sampling and end exclusions

Samples are 2 m apart within 3 km of either end and 5 m apart in between. The first 5 m at the eye (the observer’s own feet) and the last 3 m at the lamp (the car’s body) are skipped. Table S2 shows that the results have converged.

Table S2: Convergence tests on US-67 (3095 points every 30 m from the US-90 junction in Marfa toward Presidio; lidar; 0.66 m lamp; $k = 0.13$).

Setting	Road in view (km)
Reference: 2/5 m steps, 3 m exclusion at the lamp	10.44
1/2 m steps	identical point by point (3095/3095)
30 m exclusion at the lamp	10.47
60 m exclusion	10.59
120 m exclusion (discards real terrain)	11.19

S3 How robust is “visible”?

S3.1 Margins

The lamp margin d_L is the change in lamp height that puts the ray exactly on the terrain, and d_E is the same for the eye. Both follow from the clearance geometry; unit tests move the lamp or the eye

by exactly the margin and check that visibility flips. No visible point on US-67 has $|d_L| > 0.84$ m.

S3.2 Terrain-error Monte Carlo

Each elevation source gets its own, independent error field along the sight line. For a source with correlation length ℓ we generate a unit-variance first-order autoregressive sequence on a uniform grid of spacing $\Delta = \min(\ell/4, 5 \text{ m})$,

$$\delta_{j+1} = \phi \delta_j + \sqrt{1 - \phi^2} \xi_j, \quad \phi = e^{-\Delta/\ell}, \quad (\text{S3})$$

with independent standard normal ξ_j , which has correlation $e^{-|\Delta s|/\ell}$. It is interpolated linearly to the terrain samples, rescaled to unit variance (midway between grid nodes linear interpolation lowers the variance to $(1 + \phi)/2$, at worst 0.89, i.e. the standard deviation by 6%), and multiplied by the standard deviation σ of the source that supplied each sample. An independent eye-height error δ_{eye} ($\sigma = 0.15$ m) tilts the chord, adding $\delta_{\text{eye}}(D - s)/D$. In each of 200 realizations the lamp is visible if the perturbed terrain stays below the ray; P_{vis} is the fraction of realizations in which it does. Only samples that could reach the ray within 6σ are evaluated.

The error statistics of the 1/3 and 1 arc-second models were measured against the lidar at 39 807 points (`data/derived/los2/dem_error.json`): standard deviations 0.56 m and 2.05 m (biases -0.02 m and -0.09 m). Semivariograms of the difference along random transects, fitted with the exponential model $\gamma(r) = \sigma^2(1 - e^{-r/\ell})$ at lag r , with the sill fixed at σ^2 (the semivariogram of a field with correlation $e^{-r/\ell}$), give correlation lengths $\ell = 70$ m and 125 m. For the lidar we use the reported non-vegetated RMSE, 0.056 m, and an assumed correlation length of 20 m.

Classes. Visible: $P_{\text{vis}} \geq 0.95$ and still visible when 0.25 m of unresolved grass is added to the terrain. Hidden: $P_{\text{vis}} \leq 0.05$. Marginal: otherwise. Between Shafter and Marfa (64.5 km) this gives 9.5 km visible, 0.5 km marginal, and 54.5 km hidden. The earlier implementation, which classed as “visible” any point with more than 5 m of clearance far from the car, gave 9.3, 7.4, and 47.9 km; almost all of its marginal road was an artifact of coarse terrain near the car.

S3.3 Obstructions from the point cloud

On the 1-m grid of the DEM, a cell is marked obstructed (“dense” rule) when it holds at least two returns and at least 25% of its returns lie more than 0.3 m above the DEM surface; its height is raised to that of its highest return. Returns classed as noise (ASPRS classes 7 and 18) are ignored. An upper-bound variant (“max”) raises every cell to its highest return. As a check, 2.8 million ground returns differ from the DEM by a median of -0.8 mm (robust standard deviation 2.0 cm; 5th–95th percentile -4.1 to $+4.4$ cm). Only 0.8% of corridor cells carry dense cover above 0.3 m.

S3.4 Results

The “max” obstruction variant gives 9.84 km at the reference settings, and uniform clutter of 0.25, 0.5, and 1 m added to all terrain more than 10 m from both ends gives 9.36, 8.67, and 2.04 km (the 1-m case is an unrealistic bound). The limiting terrain sample lies a median 3.0 km from the car, and within 100 m of it for only 9% of points. Figure S1 shows cross-sections along four bearings.

Table S3: Road in view on US-67 (km; lidar with obstructions; Marfa to Presidio) versus lamp height and refraction coefficient.

Lamp height	$k = 0$	0.13	0.25	0.5	1
0.56 m	9.45	9.72	9.96	11.07	11.49
0.66 m (reference headlamp)	9.69	9.93	10.14	11.22	11.61
0.86 m (tail lamps)	9.90	10.14	10.32	11.52	11.88
1.12 m (high-mounted stop lamp)	10.35	10.59	10.71	11.76	12.15
1.37 m	10.59	10.83	10.98	12.09	12.42
2.5 m	11.70	11.94	12.24	13.14	13.47
4.0 m (truck clearance lamps)	13.50	13.77	14.64	14.97	15.33

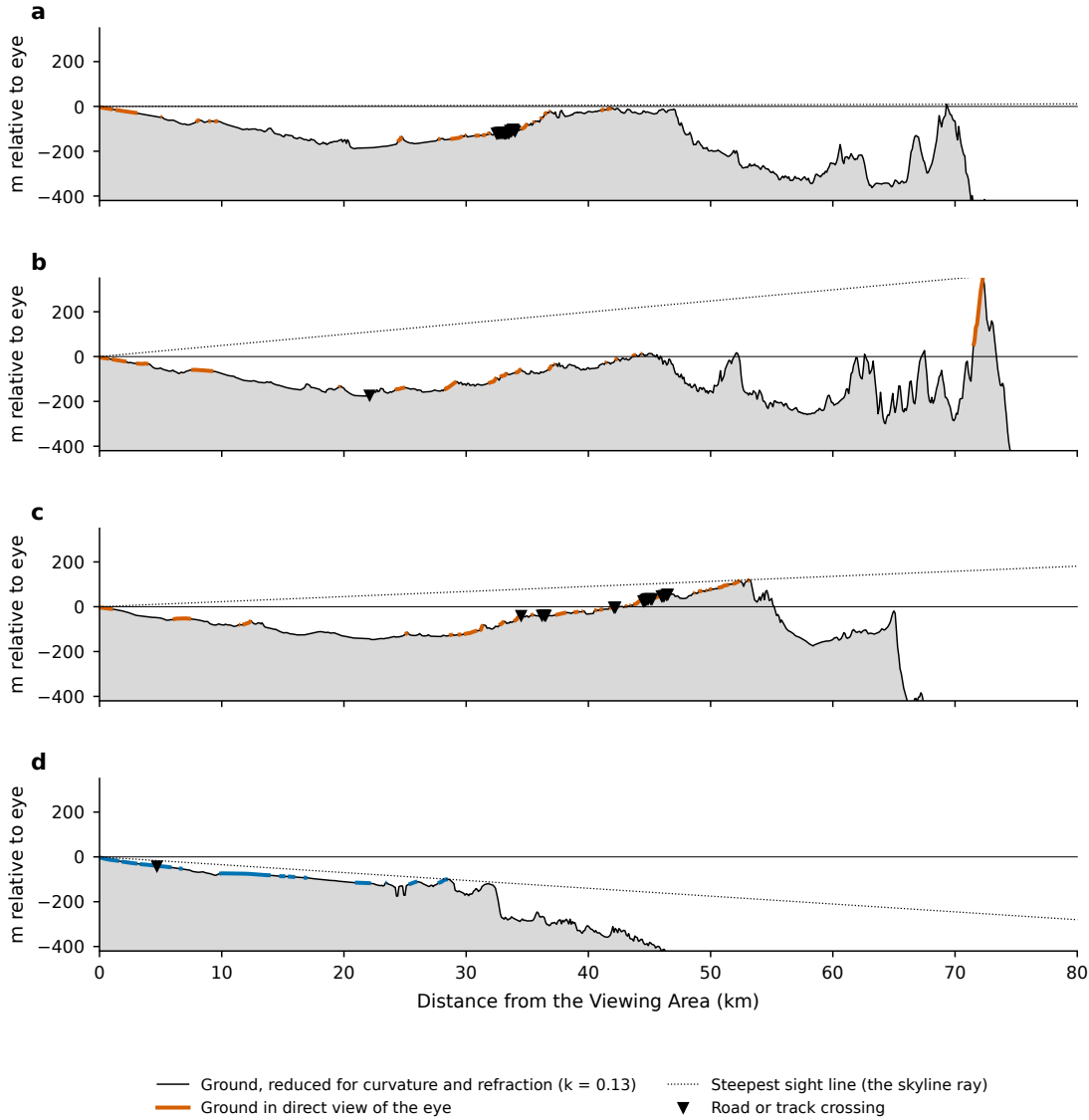


Figure S1: Terrain cross-sections from the platform, reduced for curvature and refraction ($k = 0.13$), exaggerated 18 $\times$ vertically, along (a) 233.6° true, where the view skims the far slope that carries a straight stretch of US-67 in view; (b) 240°, where a low rise on Mitchell Flat hides the highway; (c) 255.6°, across RM 2810 in the Chinati foothills; (d) 190°, along the Texas Pacifico railroad, with no highway. Colored: ground in direct view of the eye. Triangles: road or track crossings.

S4 Refraction

S4.1 Local refraction coefficient

A ray at elevation e in a horizontally stratified atmosphere curves toward the denser air with curvature $\kappa = -(1/n)(dn/dz) \cos e$. With $n - 1 = cP/T$ ($c = 79.0 \times 10^{-6}$ K/hPa at visible wavelengths) and hydrostatic pressure, $dP/dz = -g_0P/(R_dT)$ (g_0 the acceleration of gravity, R_d the gas constant of dry air), one finds, for $n \approx 1$ and $\cos e \approx 1$,

$$k = R\kappa \approx Rc \frac{P}{T^2} \left(\frac{g_0}{R_d} + \frac{dT}{dz} \right) \approx 503 \frac{P}{T^2} \left(0.0343 + \frac{dT}{dz} \right), \quad (\text{S4})$$

Eq. (4) of the main paper.[2] Here $Rc = 503$ m K/hPa for $R = 6371$ km, and $g_0/R_d = 0.0342$ K/m is the autoconvective lapse rate, which Hirt *et al.* give as 0.0343; the difference changes k by less than 0.001. At 850 hPa and 283 K the prefactor $503P/T^2$ is 5.34 m/K. (The code computes the same relation from c , g_0 , and R_d ; a unit test checks it against a numerical derivative of $n(z)$ in a hydrostatic atmosphere.)

S4.2 Ray tracing

In the chord frame the ray obeys $y'' = -k(a)/R$, where $a = y - y_{\text{ground}}(s)$ is the height above the local ground (Eq. 5); for the elevation angles here ($< 1^\circ$) the paraxial error is of relative order 10^{-4} . We launch a fan of rays from the eye, integrate each to $s = D$, discard rays that meet the terrain outside the end exclusions, and find the roots of $y(D; \theta) = 0$ by linear interpolation between neighboring rays. Each root is an image with apparent elevation equal to the chord elevation plus θ ; no root means the lamp is hidden, and more than one root is a mirage.

Profiles. Temperature follows the terrain. The stable boundary layer has the exponential form[3] $T(a) = T_0 + \Delta T(1 - e^{-a/h}) + \gamma a$, so $dT/dz = (\Delta T/h)e^{-a/h} + \gamma$. Weak: $\Delta T = 3$ K, $h = 50$ m (surface gradient 0.06 K/m); moderate: 6 K, 30 m (0.2 K/m); strong: 10 K, 20 m (0.5 K/m). Also: neutral (-0.0098 K/m), isothermal, and an elevated inversion of 4 K between 40 and 60 m above a mixed layer. These are brackets, not a climatology; Hirt *et al.* measured gradients of 1–2 K/m at 1.8 m above grass shortly after sunset, stronger than any here, but the rays of interest are within 2 m of the ground only near their two ends.

Bound. If $k \leq k_{\text{max}}$ everywhere, then $y'' \geq -k_{\text{max}}/R$ with fixed end points, so every ray lies below the constant- k_{max} arc, and a lamp with $k_{\text{crit}} > k_{\text{max}}$ is hidden. With $k_{\text{max}} = 2.82$ in our profiles, points with $k_{\text{crit}} > 2.9$ need no tracing; 212 of 1548 points (Marfa to Presidio, every 60 m) remain.

Tests. A constant gradient reproduces the constant- k arc and horizon; the reach of a surface duct matches its analytic value.

Multiple images. The elevated inversion gives three to five images at five road points (60-m samples between 18.6 and 24.4 km from Marfa), 1–8 arcseconds apart. They persist when the step falls from 5 to 1 m and the fan density rises sevenfold, but they cannot be resolved by the eye.

Table S4: Ray tracing results (US-67, 0.66 m lamp, 60-m spacing; `data/derived/los2/trace_us67.json`). k_{eq} : median constant k reproducing the traced arrival angle. Lift: rise in apparent elevation relative to $k = 0.13$. The neutral case differs from Table I of the main paper through the coarser sampling.

Profile	In view (km)	k_{eq}	Lift, median / max (arcmin)
Neutral	9.78	0.13	0 / 0
Isothermal	9.90	0.18	0.4 / 0.6
Weak stable layer	10.86	0.30	1.4 / 1.7
Moderate stable layer	10.92	0.47	2.9 / 4.3
Strong stable layer	11.28	0.67	4.5 / 7.5
Elevated inversion, 40–60 m	9.66	0.36	1.8 / 2.8

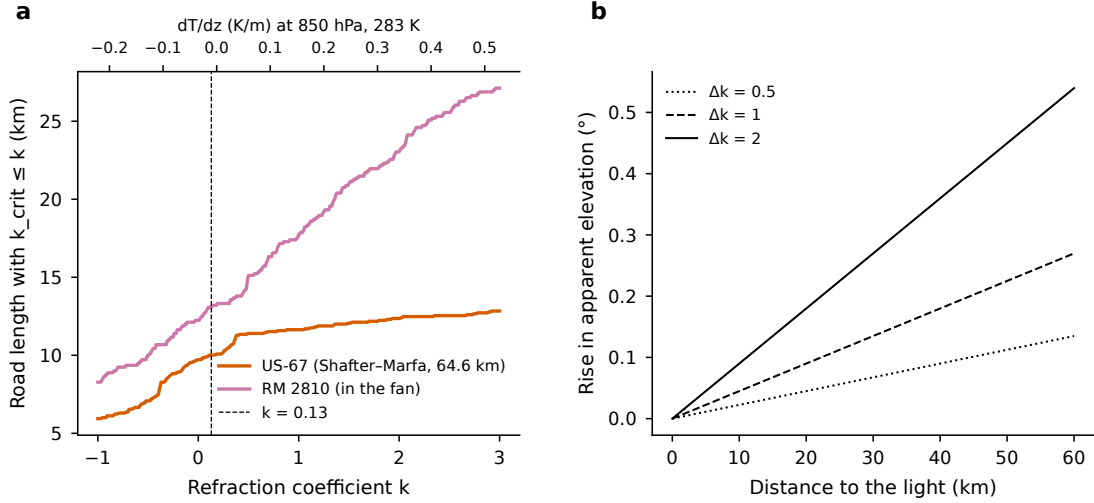


Figure S2: (a) Road in view versus refraction coefficient (constant- k model; US-67 between Shafter and Marfa, and RM 2810 within the viewing fan); the top axis gives the equivalent temperature gradient at 850 hPa and 283 K from Eq. (4). (b) Rise in apparent elevation of a light at distance d for a change Δk , $\Delta\alpha = d \Delta k / 2R$.

S5 Photometry

S5.1 Heading, grade, and view angles

The heading ψ at each 60-m road sample is the geodesic azimuth between its two neighbors along the TxDOT centerline. The grade g is the lidar height difference between points 25 m ahead and 25 m behind along ψ , divided by 50 m, and clipped to $\pm 8\%$ (no point on US-67 between Shafter and Marfa needed clipping). Both directions of travel are evaluated. The angles of the observer in the lamp frame are $h = \beta - \psi$, wrapped to $(-180^\circ, 180^\circ]$, and $v = \epsilon_Q - \arctan g - \delta_{\text{aim}}$, with $\epsilon_Q = -e_O - D/R + kD/2R$ (Eq. 6). The term $-e_O - D/R$ is the elevation of the chord at the lamp: the verticals at the eye and the lamp differ by the central angle, which D/R reproduces to better than $1''$ at these ranges. Refraction raises the ray at both ends by $kD/2R$. For equal heights and no refraction both ends see the chord at $-D/2R$. Unit tests check the formula against exact three-dimensional geometry.

On US-67 the platform lies 3° – 33° to the right of northbound drivers (5th–95th percentile; median 14°) and 0.6° below to 2.5° above their lamp axis (median $+0.8^\circ$). Southbound drivers have it 3° – 33° off their rear axis.

S5.2 Headlamp tables

The UMTRI tables[4, 5] give the 25th, 50th, and 75th percentiles of single-lamp luminous intensity at 12.8 V on a grid of 45 horizontal angles (45°L to 45°R) and 25 vertical angles (5°D to 7°U). The low-beam sample is the 20 best-selling U.S. vehicles of model year 2004 (39% of sales); the high-beam sample is the sales-weighted U.S. fleet of about 2000. The tables were read from the text layer of the published PDF reports and assigned to columns by position; every one of the 1125 cells of each table satisfies $p_{25} \leq p_{50} \leq p_{75}$, and the values at H - V (low beam 5758/8830/11 846 cd; high beam median 37 005 cd) were checked against the printed tables. Intensities are interpolated linearly in $\log_{10} I$ (floor 0.5 cd). Beyond $|h| = 45^\circ$ the intensity is either held at its 45° value or set to the regulatory minimum of the front position and side-marker lamps; the difference matters for no US-67 point in view. Two lamps are summed, and nothing is emitted forward for $|h| > 90^\circ$.

S5.3 Signal lamps

Table S5 lists the FMVSS No. 108 values used.[6] The minimum is interpolated linearly in $\log I$ between the tabulated angles, held at the last tabulated value out to 45° (the angle to which the lamps must be visible), and set to zero beyond; the maximum is applied out to 45° .

Table S5: FMVSS No. 108 photometry used for signal lamps (one lighted section; candela; horizontal angles off the lamp axis), as tabulated in NHTSA TP-108-13.

Lamp	Min., 0° – 5°	Min., 10°	Min., 20°	Max.	Number per car
Tail lamp	2.0	0.8	0.3	18	2
Stop lamp	80	40	10	300	2
High-mounted stop lamp	25	16	–	160	1

S5.4 Transmission, magnitude, and threshold

The meteorological optical range is $V = \ln 20/b$ for an extinction coefficient b ; the NPS standard visual range[7] is $3.912/b$, so $V = 0.766 \times$ the visual range. The reported 55, 90, and 165 miles become 68, 111, and 203 km. The magnitude scale of Eq. (8) follows from Schaefer’s relation[8] between point-source illuminance and magnitude: $m = 0$ corresponds to $E = 2.5 \times 10^{-6}$ lx. The Crumey threshold (Eq. 9) is his fit[9] for $20 < \mu < 22$ mag arcsec $^{-2}$; the field factor F collects the observer’s acuity, experience, and conditions. Table S6 gives m_{lim} over the ranges used.

Table S6: Naked-eye limiting magnitude from Eq. (9).

μ (mag arcsec $^{-2}$)	$F = 1.4$	2.0	2.4	4.0
20.5	6.05	5.67	5.47	4.91
21.0	6.25	5.86	5.66	5.11
21.5	6.44	6.05	5.85	5.30
22.0	6.63	6.24	6.04	5.49

S5.5 Further results

Where a northbound car’s headlamps are in view, the low-beam gradient $d \log_{10} I/dv$ at the observer’s position is -0.25 per degree (median; interquartile -0.14 to -0.37), a factor of 1.4–2.3 per degree. Within one window longer than a single sample the brightness changes by a median 1.1 magnitudes (maximum 4.9); between successive 60-m samples (2 s at 30 m/s) by more than 0.8 magnitudes one time in ten (maximum 2.7). On RM 2810 only the direction toward Marfa faces the platform ($|h|$ from 3° to 42° , median 18°): low-beam median $m = 4.0$, brightest -0.7 ; high beam brightest -2.4 .

S6 One car and occupancy

A window is a maximal run of consecutive 60-m samples in view ($P_{\text{vis}} \geq 0.5$ for the lamp’s height); its duration at speed u is its length divided by u . Windows shorter than 60 m cannot be resolved. For vehicles arriving at random with one-way flow q , the number in a road section of length L at speed u is Poisson with mean qL/u (Eq. 10), and the probability that at least one is in view is $1 - e^{-qL/u}$. Each vehicle produces one appearance per window, so a flow q produces q times 18 appearances per hour on US-67.

For the occupancy we use the 2025 AADT (1121 and 1383 vehicles per day at stations 189D5 and 189H5A, both directions), a daily-mean one-way flow of 23–29 vehicles per hour. The rate map of Sec. S7 uses, more conservatively, the highest of the three most recent annual counts (2331 and 1549). No hourly night-time count exists for this road; night flows are lower than the daily mean, and a traffic count during an observing session should replace these values.

S7 Known-source catalogue, mask, and rate map

S7.1 Catalogue

The catalogue contains: US-67 (60-m samples) and the state roads in the viewing fan (RM 2810, US-90, US-67/90 beside the platform, FM 170, SH 118, RM 169, FM 1112, SH 17; 120-m samples; lamp

0.66 m) and the one county road with a stretch in view inside the fan, Nopal Road (CR 189-0002; from the lidar line-of-sight run, thinned to 120 m); the Union Pacific and Texas Pacifico railroads (locomotive headlight at 4 m); every FCC-registered tower with registered lighting (top light; 29 of 65 registered structures in the area are lit, and 7 of those show their top light at $k = 0.13$); the towns of Marfa, Shafter, Presidio, and Ojinaga (direct light where in view for some $k \leq 1$, otherwise a skyglow band from the skyline to 0.5° above it over $\pm 1^\circ$ of bearing); and the tethered aerostat at 292.9° (a full-height column, because its altitude varies).

S7.2 Known-source mask

Each source i with $k_{\text{crit}} \leq 1$ occupies the vertical segment of apparent elevation from $\alpha_i(\max(k_{\text{crit}}, 0))$ to $\alpha_i(1)$, where $\alpha_i(k) = \alpha_i(0.13) + d_i(k - 0.13)/2R$. Neighboring samples of the same road or track less than 1 km apart are joined. The union is widened by the observer’s pointing error, $\pm 0.3^\circ$ in bearing and $\pm 0.1^\circ$ in elevation for photographic pointing (a skyline-referenced photograph or a theodolite) and $\pm 3^\circ$ by $\pm 0.25^\circ$ for compass pointing, rasterized at $0.02^\circ \times 0.04$ mrad, and converted to polygons (`data/derived/zos.json`). Within the 120° fan, in the band 5 mrad deep below the skyline, the mask covers 33% at photographic and 74% at compass pointing. Towns beyond the terrain model (Presidio and Ojinaga) are treated as skyglow.

S7.3 Rate map

The expected number of catalogued transient lights per hour in each patch of the view is

$$\lambda = \sum_s q_s P_{\text{det},s}, \quad (\text{S5})$$

where for a road $q_s = \text{AADT} \times 0.02/2$ per direction (2% of daily traffic per night hour; no measured night share was found) and $P_{\text{det}} = p_{\text{high}}[m_{\text{high}} \leq m_{\text{lim}}] + (1 - p_{\text{high}})[m_{\text{low}} \leq m_{\text{lim}}]$ with $p_{\text{high}} = 0.25$ (high-beam use on isolated rural roads[10]), $m_{\text{lim}} = 5.86$, and $V = 111$ km. For a railroad q is the largest number of night-time through trains reported at any of its crossings divided by 12 h. A road contributes its rate once per patch; rates are the maximum along one line and the sum across lines. Permanent lights (towers, towns, the aerostat) are a separate “fixed light” layer.

In the band 5 mrad below the skyline, the fraction of the view with $\lambda < 0.01 \text{ h}^{-1}$ and no fixed light (“quiet”) is 0.768 on a standard night ($k = 0.13$), 0.732 with strong refraction ($k = 1$), and 0.760 if southbound tail lamps at the FMVSS maximum are counted. With compass pointing it is 0.360. Nopal Road has no traffic count; it is given an assumed AADT of 60, the order of the least-travelled station on RM 2810 (20–52). Figure S3 shows the map (`data/derived/weighted_zone.json`).

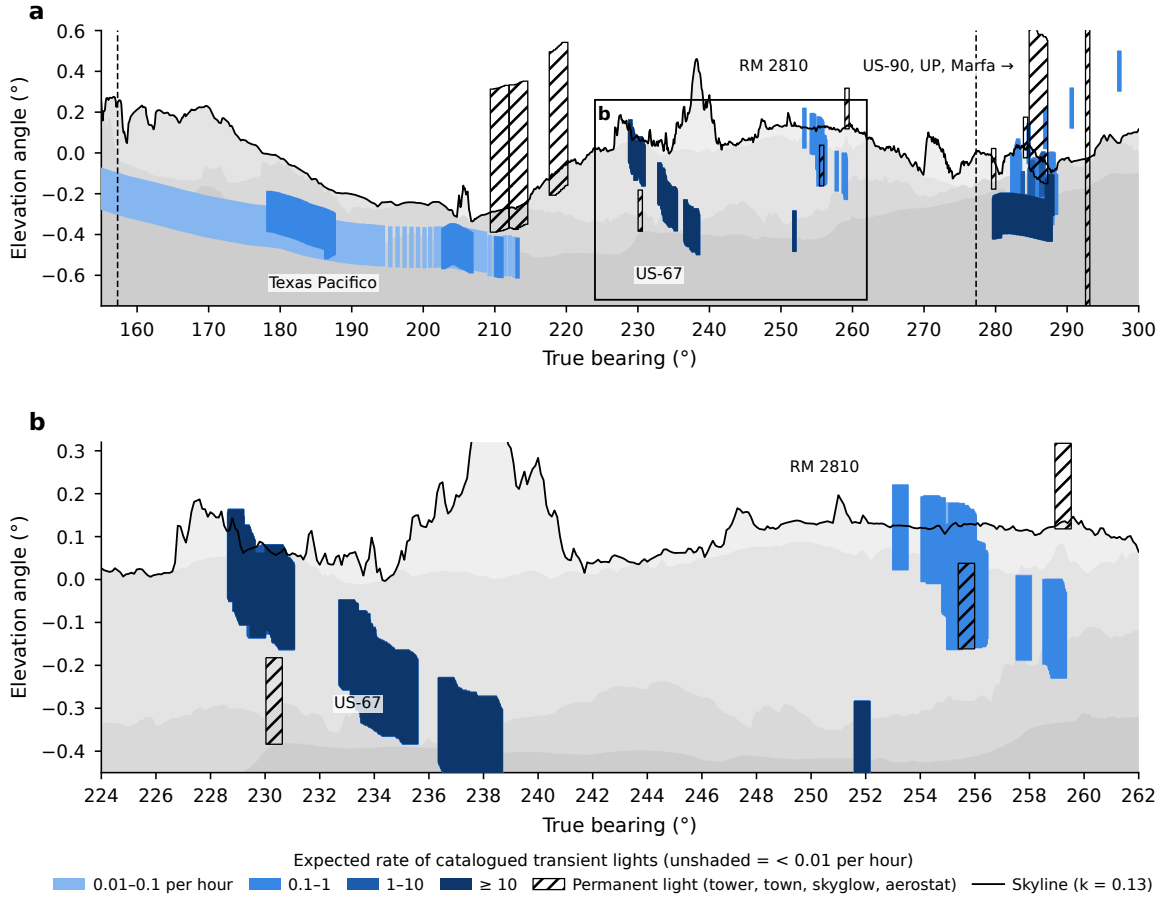


Figure S3: Expected rate of catalogued transient lights (per hour) across the view on a standard night ($k = 0.13$, $V = 111$ km, pointing error $\pm 0.3^\circ \times \pm 0.1^\circ$), with fixed lights hatched. (a) 155° – 300° , vertical scale exaggerated $29\times$; the box is the region of (b). (b) The US-67 and RM 2810 sector, 224° – 262° , exaggerated $16\times$. Northbound US-67 traffic reaches about 23 lights per hour and RM 2810 at most 0.5; the Texas Pacifico track carries at most 0.08 trains per hour.

S8 Photograph analysis

Data. Ten frames taken from the platform on 21 November 2018 with a Sony ILCE-6300 (APS-C, 23.5 mm sensor, 6000 px wide) and an E 55–210 mm lens at 210 mm (nominal scale 936 px per degree), exposures 1/25 s to 45 s. A sunset frame from the same camera calibrates its clock.

Registration. In each frame the sky–land boundary is traced where red minus blue drops sharply. A rectilinear camera model with four parameters (azimuth A_0 , elevation E_0 , roll, focal-length scale) is fitted to the modeled skyline ($k = 0.13$) by robust least squares. Residuals are 0.005–0.017°. The fitted scale is 0.974–0.977 in every frame, an effective focal length near 205 mm. The best alternative pointing fits 1.4 to 10 times worse. Matching the solar disc in the sunset frame to its computed position (Astronomy Engine,[11] standard refraction) shows the camera clock 59 min fast, so the frames were taken at 18:12–18:26 CST, 16–30 min after sunset (17:56 CST).

Detection. A high-pass image (the frame minus its 21×21 px median) is thresholded at 7 robust standard deviations; detections must have 4–600 px and lie between the modeled 10-km ridge line and the skyline. A contrast rule (peak/median ≥ 1.20) was set after inspecting the detections; results are reported with and without it and do not change.

Independent sightings. Frames seconds apart show the same vehicles, so the unit of analysis is a sighting within a burst (frames 04+05, 06+07, 08, 09, 10). Two detections in one burst are the same vehicle if they lie on the same road and their along-road separation is within what a vehicle at ≤ 45 m/s covers between frames, plus 60 m. This gives eight sightings.

Statistic and null model. The statistic is the median angular distance from the sightings to the modeled curve of visible road (piecewise linear between consecutive in-view 60-m samples). The null places the same number of random points per burst uniformly in each frame’s ground band. The observed median is 0.0085° (maximum 0.013°), against a null median of medians of 0.28°; only 1.4% of the band lies within 0.0085° of visible road. The exact probability that at least 4 of 8 random points fall that close (Poisson-binomial, with each burst’s own fraction) is 3×10^{-6} ; because a median of 8 values can only be as small as x if at least four values are, this is an upper bound on the chance of a median as small as observed (2×10^{-6} without the contrast rule); no one of 4000 Monte Carlo draws was as close.

Visible–hidden boundary. 48–50% of the US-67 samples that project into each frame are modeled as hidden, yet the road sample nearest each of the eight sightings is a visible one. If the model’s split had no predictive power, the probability would be the product of the frames’ visible fractions, about 0.005.

Speeds. Frames 04→05 (6 s): 179 ± 30 m along the road, 30 m/s (17–48 m/s at 2σ), including 1-s timestamp resolution and a 0.006° position error. Frames 06→07 (2 s): 41 m/s (16–114 m/s); a second pair is unconstrained because the road points almost at the platform there.

Long exposure. In the 45-s frame, streaks are found without the model (7 robust standard deviations, below the skyline, width ≥ 60 px, aspect ratio ≥ 5). Streak 1 (236.90°–237.36°) lies a median 0.0049° from visible road; streak 2 (233.93°–234.02°, 88 px) lies 0.028° from it and is of undetermined origin.

Limitations. Because the pointing is fitted to the modeled skyline, these tests measure internal consistency, not absolute pointing. The frames are not photometrically calibrated and do not constrain k , which is degenerate with E_0 .

S9 Verification

Forty-five unit tests (`analysis/tests/`) pass:

- *Geometry*: Euler normal-section radius; smooth-Earth horizon distance for $k = 0, 0.13, 0.5$; dip of a sea-level horizon; k_{crit} against brute force on random terrain; an isolated ridge against its analytic k_{crit} ; the multi-lamp engine against the single-lamp engine; lamp and eye margins flip visibility; fine sampling near both ends; missing terrain raises an error; clutter raises k_{crit} .
- *Monte Carlo*: statistics of the correlated error field; the deterministic limit; the analytic probability for a pure eye-height error.
- *Ray tracer*: Eq. (4) gives the standard value; constant gradients reproduce constant- k arcs and horizons; the reach of a surface duct matches its analytic value.
- *Equations* (`test.equations.py`, independent of the model code): the curvature drop and horizon against an exact sphere; the parabolic sag against the exact circular arc; k_{crit} as the visibility threshold; the refraction coefficient of Eq. (4) against a numerical derivative of $n(z)$ in a hydrostatic atmosphere; the comparison bound for height-dependent k ; the elevation at the lamp, Eq. (6), against exact three-dimensional geometry; the pitch sign; optical and visual range; the magnitude constant (from Schaefer’s foot-candle form, and the Sun as a check); the Crumey values; the occupancy, Eq. (10), against a simulation of random traffic; the angular drift rate; Euler’s radius; the autoregressive error field; and the conservativeness of the median test.
- *Photometry*: tabulated UMTRI values; the magnitude scale; the definition of optical range; the Crumey relation; sign conventions of h and v ; level-ground symmetry; signal-lamp brackets.

An earlier, independent implementation (1 arc-second terrain, 15-m step, 30/60-m end exclusions, spherical Earth) is reproduced when run with its settings: visibility agrees for 1075 of 1077 points, and its “clearance” equals our lamp margin to -0.03 m (median). Its azimuths were 0.117° too large on average (0.145° at most) and its distances off by up to 72 m, from mixing spherical and ellipsoidal geometry. The photograph registration absorbed exactly this azimuth offset in the earlier analysis.

S10 The companion website

The model is published as a website, <https://zacharyslate.github.io/marfa-lights-investigation/>, built from the same derived data. It works on phones and offline once loaded. Its pages:

- *Map and light identifier*: a map and panorama of the view with every catalogued source (the US-67 line of sight, refraction, railroads, airfields, towers, and towns). The observer enters the bearing and elevation of a light, uses the phone compass, or taps the map or panorama, and the tool lists every known source on that line within a chosen window and, for roads, how bright a car there would look.
- *Sky finder*: points the phone camera at the view and draws the skyline, US-67, RM 2810, towers, and stars on the live image; after calibration on a known light it records a calibrated bearing and elevation.
- *Report a sighting*: a form that checks a reported light against every known source and keeps the report on the user’s device until the user chooses to send it.
- *Sight lines*: where US-67 can be seen from the platform, point by point.
- *Science*: the figures of this work with plain-language explanations.

S11 Reproducing the results

From the repository root, with the Python environment described in the repository:

```
cd analysis
python -m marfa.fetch_dem --download      # terrain, point cloud, geoid
python -m marfa.fetch_dem --verify        # check against MANIFEST.json
python -m marfa.run_los us67 --dem lidar  # line of sight, US-67
python -m marfa.run_los roads --dem lidar --spacing 60
python -m marfa.run_trace                  # ray tracing
python -m marfa.run_photometry
python -m marfa.run_traffic
python -m marfa.export_site
cd ..
python analysis/headlight_model.py
python analysis/build_site_data.py
python analysis/zone_of_skepticism.py     # known-source mask
python analysis/weighted_zone.py          # rate map
python analysis/pub_figures.py
python analysis/photo_figures.py
python analysis/manuscript_figures.py
pytest analysis/tests
```

The exact options of each step are given in `publication/README.md`.

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